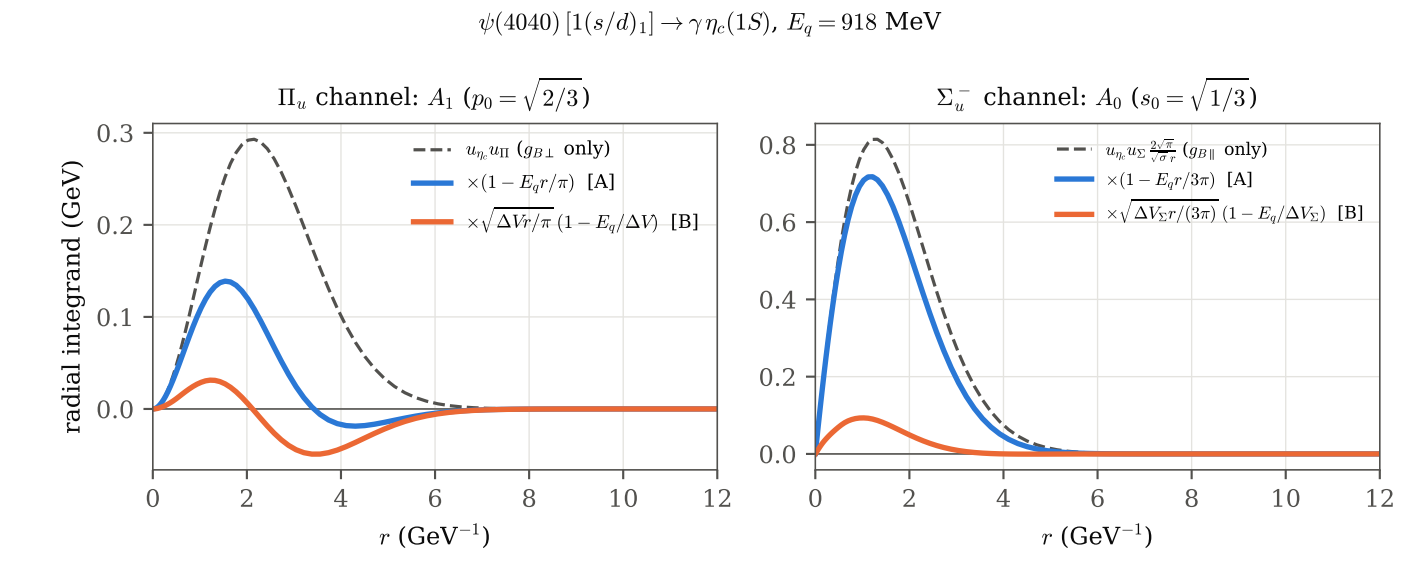
$\psi(4040)\,[1(s/d)_1] \to \gamma\,\eta_c(1S)$, $E_q = 918$ MeV
Π_u channel: A_1 ($p_0 = \sqrt{2/3}$)
$u_{\eta_c} u_\Pi$ ($g_{B\perp}$ only)
$\times(1 - E_q r/\pi)$ [A]
$\times\sqrt{\Delta V r/\pi}\,(1 - E_q/\Delta V)$ [B]
radial integrand (GeV)
r (GeV^{-1})
Σ_u^- channel: A_0 ($s_0 = \sqrt{1/3}$)
$u_{\eta_c} u_\Sigma \frac{2\sqrt{\pi}}{\sqrt{\sigma}\, r}$ ($g_{B\parallel}$ only)
$\times(1 - E_q r/3\pi)$ [A]
$\times\sqrt{\Delta V_\Sigma r/(3\pi)}\,(1 - E_q/\Delta V_\Sigma)$ [B]
r (GeV^{-1})